Gauge-Invariant Residue Cocycles and a Scalar Renormalization Obstruction

Physical Weights Along the Two Conjugate Branches

Liang Wang

July 2026

Abstract

The branch correspondence of RH-204 permits a gauge-free comparison of source–observation residues across scales. For matched nonzero residues $r_{c,j}, r_{f,j}$, the exact diagonal multiplier is $m_j = r_{f,j}/r_{c,j}$. These multipliers compose under refinement and occur in conjugate pairs for real physical data.

We test whether the four multipliers can be replaced by one scale-dependent complex scalar. The least-squares optimum is explicit. On the four physical transitions its relative residual ranges from 0.32371 to 0.99985; on the first left transition the optimal scalar is nearly zero while the two branch multipliers have moduli 0.10961 and 0.15002 with different phases. The exact diagonal cocycle residual is below 1.29×10^{-16}, and conjugation symmetry holds within 3.23×10^{-14}.

Thus a common scalar normalization cannot transport the physical transfer weights, even though a branch-diagonal endpoint cocycle always can. The cocycle is source/observation dependent and differs between left and right channels by as much as 1.05364; it is therefore not yet an intrinsic spectral renormalization. The negative result motivates separating the weighted transfer ledger from the unweighted quartet divisor.

1 Residues after branch matching

For a simple eigenvalue λ with right and left vectors v, w chosen so that $w^*v = 1$, the spectral projector and physical residue are

$$P_\lambda = vw^*, \qquad r_\lambda = \operatorname{tr}(OP_\lambda S). \tag{1}$$

RH-195 proves that this is also the Frobenius pairing of the source and observation channel states [1].

RH-204 supplies the matched labels $j \in \{-, +\}$ for the two upper-half-plane branches and their conjugates [2]. No eigenvector phase is used in this labeling.

2 Gauge invariance

Proposition 2.1 (Gauge invariance)**.** *Under the biorthogonal rescaling*

$$v \mapsto \alpha v, \qquad w \mapsto \overline{\alpha}^{-1} w, \qquad \alpha \neq 0,$$

both P_λ and r_λ are unchanged.

Proof. The rescaled outer product is $(\alpha v)(\overline{\alpha}^{-1} w)^* = vw^*$. Substitute this identity into (1). □

Consequently, the large residue changes in RH-202 cannot be removed by rephasing or balancing individual eigenvectors.

3 Diagonal residue cocycle

For every matched nonzero residue define

$$m_j(c, f) = \frac{r_{f,j}}{r_{c,j}}. \tag{2}$$

Theorem 3.1 (Finite cocycle law)**.** *For three levels a, b, c on one consistently labeled branch,*

$$m_j(a, c) = m_j(b, c) m_j(a, b). \tag{3}$$

If the operators, source, and observation are real, then

$$m_{\overline{j}}(a, b) = \overline{m_j(a, b)}. \tag{4}$$

Proof. Equation (3) is cancellation of the intermediate nonzero residue. Reality gives $r_{\overline{\lambda}} = \overline{r_\lambda}$, and taking ratios gives (4). □

The diagonal cocycle is exact but endpoint dependent. It does not predict a new residue from coarse data alone.

4 Best common scalar

Let $r_c, r_f \in \mathbb{C}^4$ collect the ordered quartet residues. Consider

$$\min_{\alpha \in \mathbb{C}} \|r_f - \alpha r_c\|_2. \tag{5}$$

Proposition 4.1 (Scalar optimum)**.** *If $r_c \neq 0$, the unique minimizer is*

$$\alpha_* = \frac{r_c^* r_f}{r_c^* r_c}. \tag{6}$$

The normalized residual vanishes if and only if r_f is a scalar multiple of r_c, equivalently all nonzero branch multipliers agree.

This is a strict, basis-free test of scalar scale renormalization. A large residual means that no global amplitude and phase correction can align all physical weights.

5 Physical multiplier table

The upper-half-plane branch multipliers and scalar residuals are:

step	side	m_-	m_+	scalar residual
$0.04 \to 0.02$	left	$0.1037 + 0.0356i$	$-0.0269 - 0.1476i$	0.99985
$0.04 \to 0.02$	right	$0.1009 + 0.1534i$	$0.0372 - 0.1966i$	0.97654
$0.02 \to 0.01$	left	$1.5204 - 0.3314i$	$1.3920 + 0.6344i$	0.39406
$0.02 \to 0.01$	right	$0.4963 - 0.0836i$	$0.9778 + 0.3073i$	0.32371

The first transition is nearly orthogonal to a common-scalar description. Even on the finer transition, one scalar leaves a 32–39% residual.

The diagonal reconstruction

$$r_f = \mathrm{diag}(m_1, \dots, m_4) r_c$$

has maximum relative residual 1.2846×10^{-16}, as required by the exact definition. Conjugate multiplier errors are below 3.23×10^{-14}.

6 Channel dependence

At $0.04 \to 0.02$, the largest difference between corresponding left and right branch multipliers is 0.11781. At $0.02 \to 0.01$ it is 1.05364. Therefore the multiplier is not currently universal across the two physical realizations.

Across both transitions, the composed upper-branch multipliers are

side	$m_-(0.04, 0.01)$	$m_+(0.04, 0.01)$
left	$0.16942 + 0.01977i$	$0.05617 - 0.22251i$
right	$0.06290 + 0.06770i$	$0.09681 - 0.18076i$

The telescoping law is exact, but the two-step numbers reinforce rather than remove branch and channel dependence.

This is expected from (1): changing S or O changes the weights without changing the eigenvalue divisor. A transfer determinant may need this cocycle, while an unweighted spectral determinant should not.

7 Three distinct notions of renormalization

The audit separates:

1. *eigenvector gauge*, which leaves residues unchanged;
2. *common scalar scale normalization*, rejected in the four cases;
3. *branch-diagonal endpoint cocycle*, exact but source dependent.

Conflating these notions would hide the obstruction by fitting one multiplier per datum.

8 Consequence for determinant design

The weighted moments

$$h_q = \sum_j r_j \lambda_j^q \tag{7}$$

carry the residue cocycle. The Newton traces

$$s_q = \sum_j \lambda_j^q \tag{8}$$

do not. RH-199 already proves that these two ledgers must not be identified. The present obstruction strengthens the practical reason to study the unweighted quartic divisor separately before attempting a physical feedback factor.

9 Claim boundary and next step

The gauge and cocycle statements are exact finite algebra. The rejection of a common scalar is a four-case floating result, not an all-level theorem. No residue lower bound, source-independent cocycle, von Mangoldt weight, or arithmetic trace identity is obtained. RH-207 next studies the unweighted quartic divisor across channels and scales.

10 What a useful residue theorem would require

An all-level weighted transfer ledger would need more than nonzero endpoint residues. One needs a uniform lower bound preventing loss of observability, an upper bound on cocycle products, and compatibility with the growing-cloud selection. If multipliers are allowed to be fitted independently at every branch and level, exact reconstruction is tautological and provides no compactness. A useful theorem must derive them from source/observation regularity and operator dynamics.

References

[1] Liang Wang. Source–observation riesz channels, 2026. RH-195 preprint.

[2] Liang Wang. Conjugate-branch correspondence across small-noise levels, 2026. RH-204 preprint.